

Robot HAT Hardware and Robotics Applications

Name: Mahdi Muhamadkhani Course: ITAI 2374 Date: September 27, 2026

Introduction

Robot HAT is a Raspberry Pi expansion board that connects software to physical sensors and actuators. This report focuses on Robot HAT V5 in the PiDog V2 kit, which supports Raspberry Pi 5. Its onboard microcontroller handles PWM generation and analog conversion while the Pi runs movement planning, vision, and AI software. The board simplifies wiring and supplies interfaces that a robotics project would otherwise need to assemble separately. [1, 2]

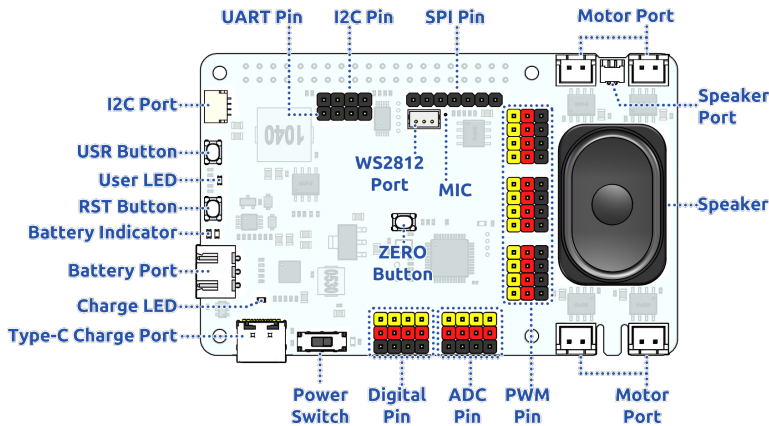


Figure 1. Robot HAT V5 ports and controls. Source: SunFounder [4]. Camera CSI connection is on the Raspberry Pi.

Key Hardware Components and Functions

Component	Function and practical application
Servo ports and MCU	Twelve PWM outputs, P0-P11, provide servo control signals. Pulse width determines commanded joint position. The Pi communicates with the onboard MCU over I ² C, allowing coordinated movements without generating every pulse in the main application. [2]
DC motor drivers	Four motor channels control speed and direction for wheeled robots. V5 lists 5 V and up to 1.8 A per motor channel. PiDog uses positional servos for its joints, leaving these DC outputs unused. [3]
ADC and sensor inputs	Four external 12-bit ADC inputs, A0-A3, convert analog sensor voltages into values from 0 to 4095. A separate internal channel measures battery voltage. Digital connections support ultrasonic trigger and echo signals for distance sensing. [2, 4, 5]
Expansion and camera	I ² C connects compatible sensors; UART supports serial devices; SPI supports suitable peripherals. The WS2812 lighting connection shares SPI MOSI. A camera connects directly to the Raspberry Pi CSI interface, not to a camera socket on the HAT. [4, 6]
Audio and indicators	I ² S audio provides speaker output and microphone input for spoken responses and voice interaction. Audio conversion is distinct from the sensor ADC; no general-purpose DAC output is documented. Power, charging, battery, and user LEDs help diagnose operating state. [3, 4]
Power management	A 6.0-8.4 V, two-cell battery pack connects through the three-pin battery input. USB-C charges the pack. Regulation supplies the Pi and peripherals; charging balance and protection circuitry support battery operation. The switch controls board power. [3, 7]

Example Use Case

In a proposed obstacle-avoiding PiDog application, eight leg servos produce walking motions, three position the head, and one moves the tail. Software reads ultrasonic distance and pauses or redirects the walking sequence when an obstacle is too close. Camera images can support object recognition, while the speaker announces a response. The HAT provides sensing and actuation interfaces; the Pi executes the decision-making software. This links sensor measurements to programmed physical behavior. [1, 5, 6, 8]

Challenges and Considerations

Check the exact board and kit revision: the older PiDog Standard kit is not the Pi 5-compatible V2 kit. Budget current for the Pi and simultaneous servo loads; motor channel ratings do not increase the shared supply capacity. Confirm connector polarity, signal voltage, and common ground before powering up. Use the correct Pi 5 camera cable, calibrate servo centers, and avoid conflicting shared pins or PWM timer settings. Install the matching Robot HAT library and audio configuration. [1-4, 6]

References

SunFounder documentation, accessed September 27, 2026. 1 PiDog versions · 2 MCU and PWM · 3 V5 specifications · 4 V5 hardware and figure · 5 Ultrasonic example · 6 Camera · 7 Battery · 8 PiDog servo groups